AISSMS
COLLEGE OF ENGINEERING
ज्ञानम् सकलजनहिताय
Accredited by NAAC with "A+" Grade

BE-E&TC FINAL YEAR PROJECT AY(2025-2026)

Project Title:

SUSTAINABLE HYDROPONIC FARMING INTELLIGENT AUTOMATION AND MONITORING

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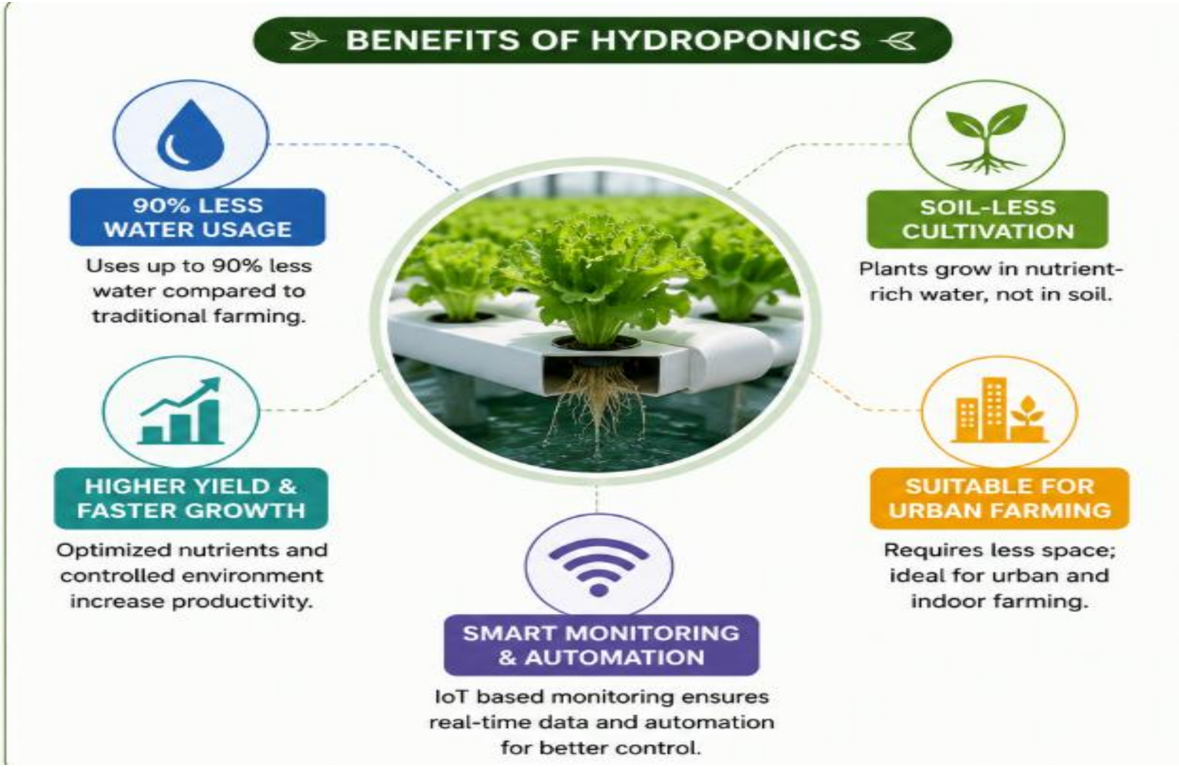
Guide: Prof Dr. R. R. Itkarkar

Introduction

- Hydroponics is a soil-less farming technique where plants grow in nutrient-rich water.
- It enables efficient utilization of water and nutrients.
- Continuous monitoring is required for healthy plant growth.
- IoT technology enables real-time monitoring and automation.

Need of Hydroponics



Problem Statement

Manual monitoring and control of hydroponic systems are inefficient, error-prone, and lack real-time decision-making, leading to suboptimal plant growth and resource wastage.

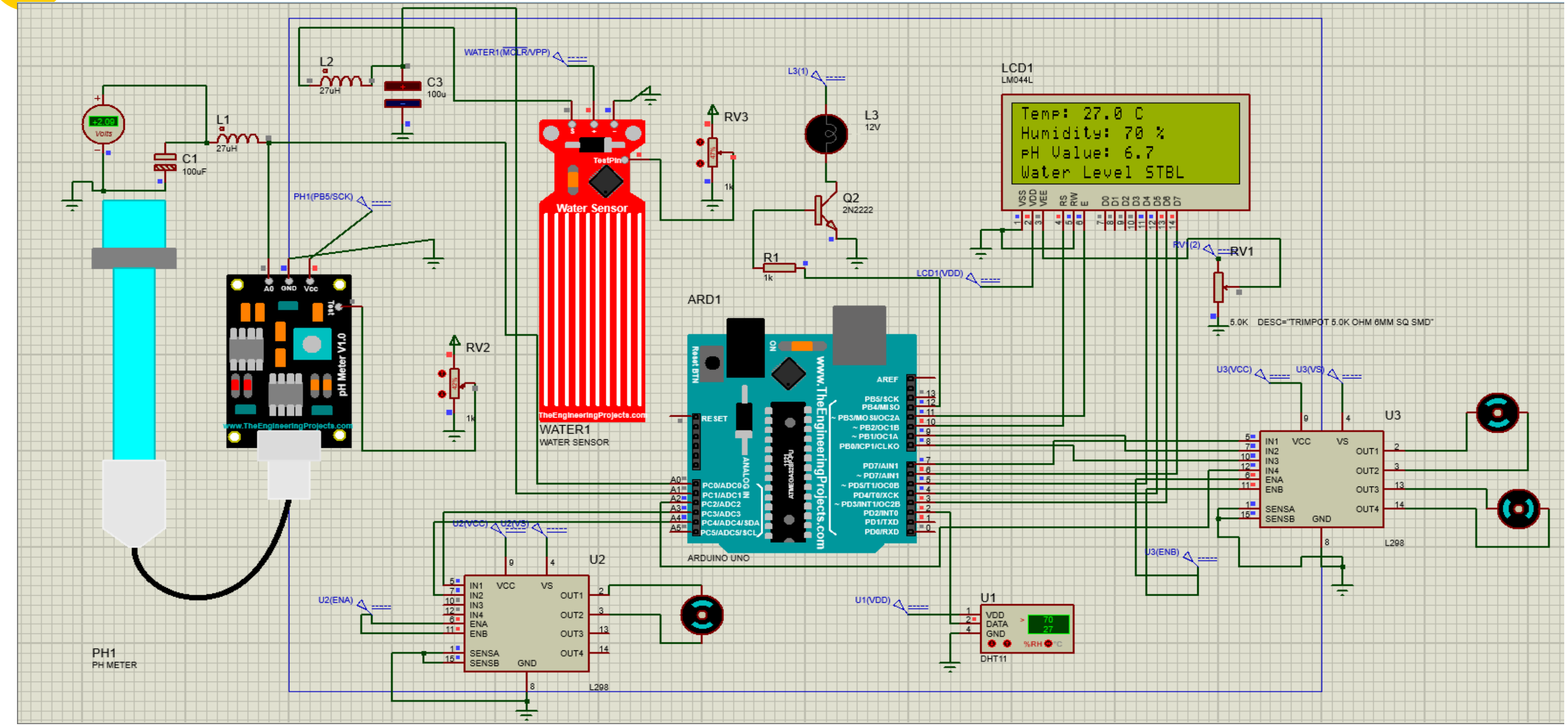
Aim:

To design and develop an IoT-based hydroponic farming system for real-time monitoring and control of plant growth parameters.

Objective:

- To implement real-time monitoring of pH, temperature, humidity, and water level.
- To automate water flow using pumps.
- To develop a system capable of assisting urban farming and sustainable agriculture.
- To provide wireless visualization of system data using Blynk IoT and OLED display.
- To generate real-time notification and email alerts when water level falls below threshold.

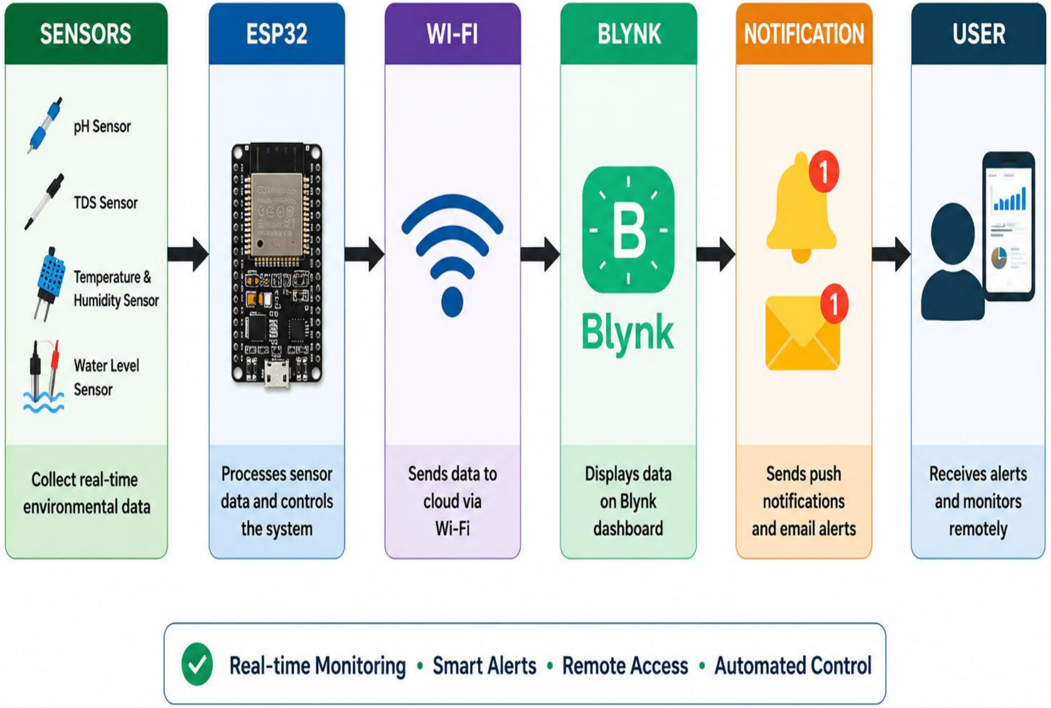
Simulation



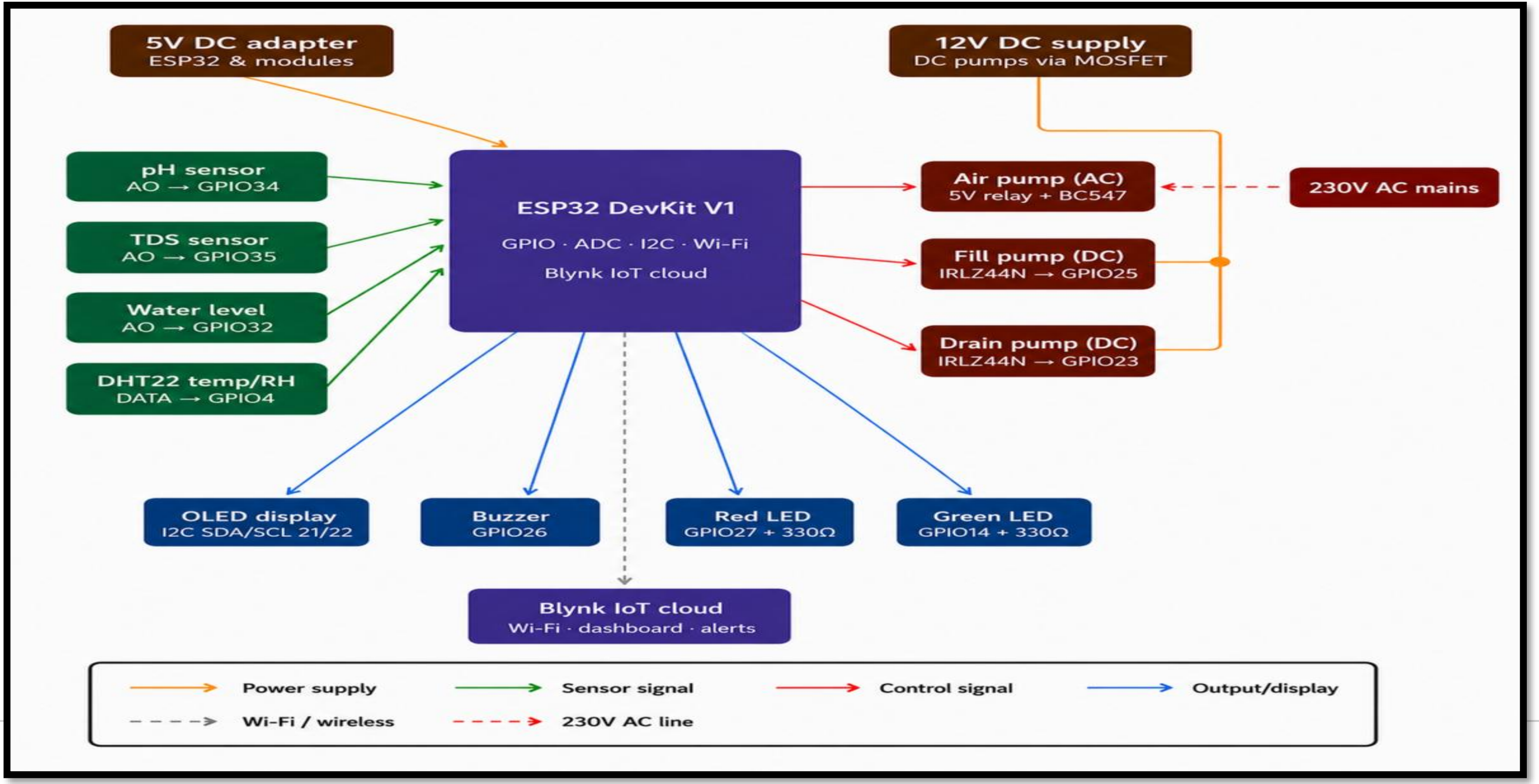
Temp: 27.0 C
Humidity: 70 %
pH Value: 6.7
Water Level STBL
LCD1
LM044L
ARD1
ARDUINO UNO
WATER1
WATER SENSOR
Water Sensor
PH1
PH METER
pH Meter V1.0
U1
DHT11
U2
L298
U3
L298
RV1
RV2
RV3
L1
L2
L3
C1
C3
Q2
2N2222
R1
5.0K DESC="TRIMPOT 5.0K OHM 6MM SQ SMD"

Proposed Solution

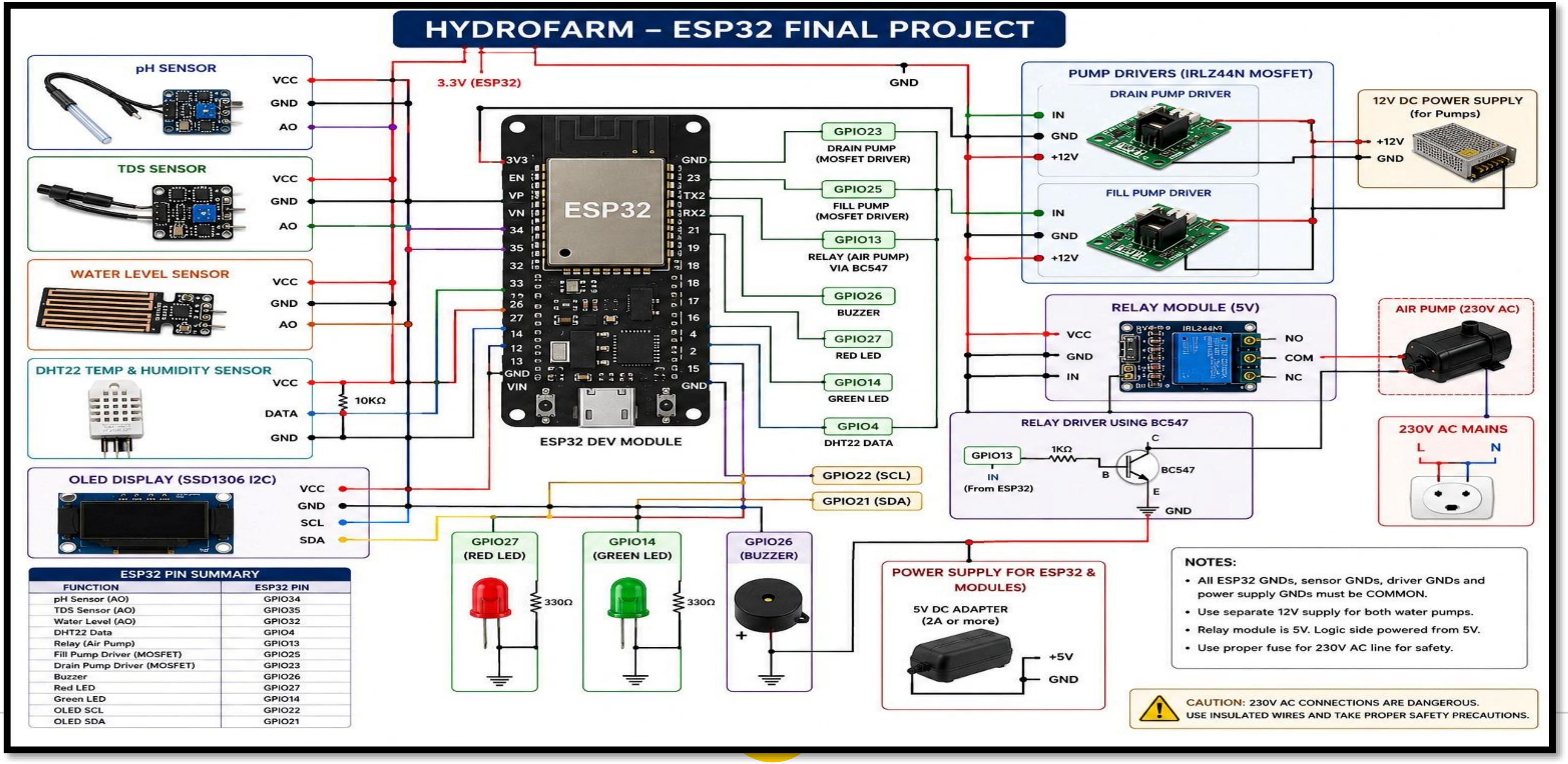
- ESP32-based intelligent monitoring system.
- Integration of multiple environmental sensors.
- Real-time monitoring through Blynk IoT.
- Automated water level management.
- Push notifications and email alerts.
- Local display using OLED.



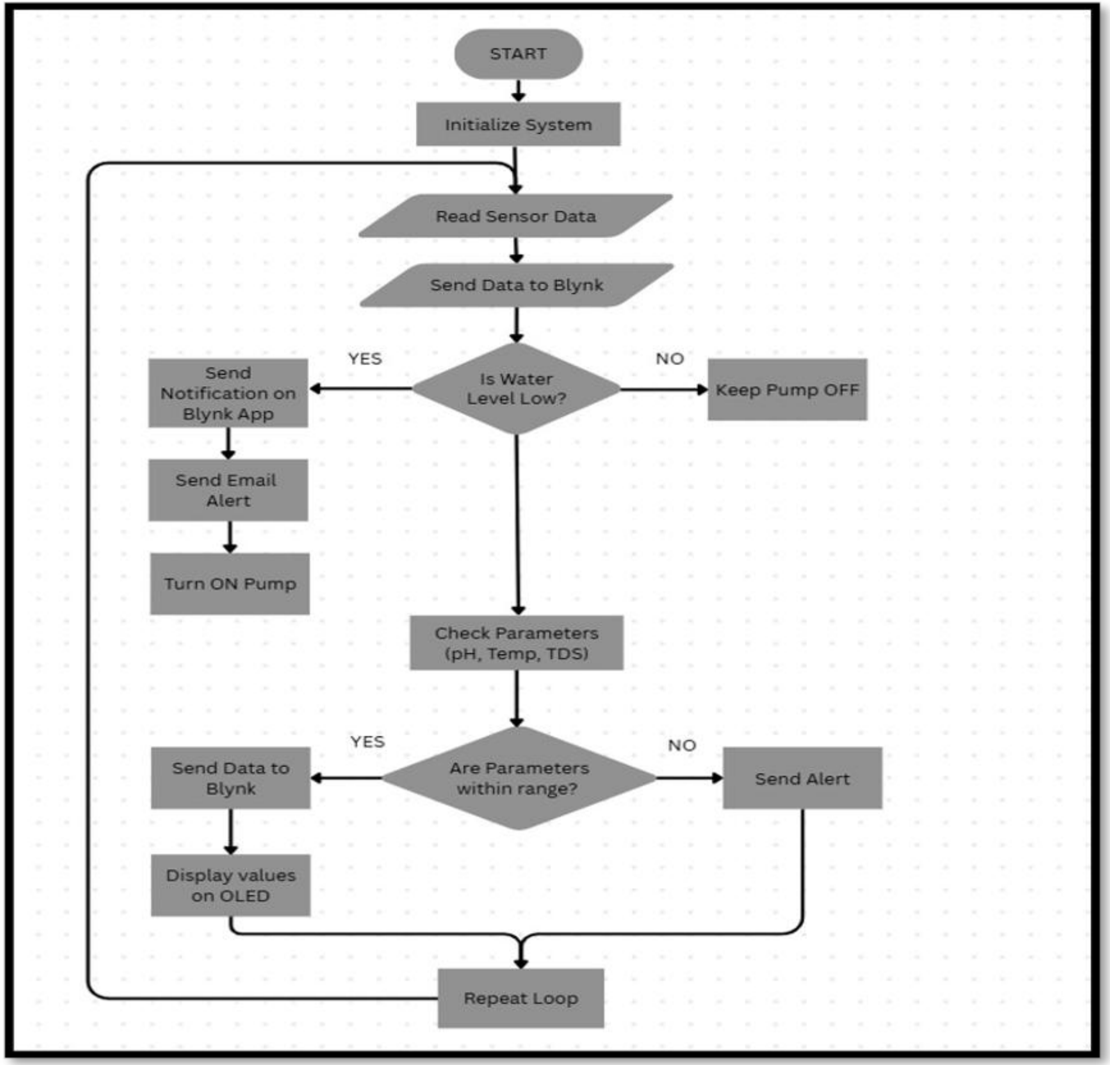
Block Diagram:



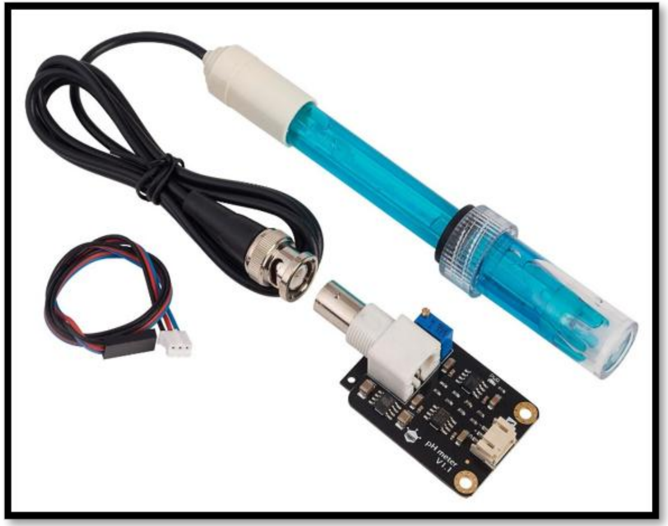
Circuit Diagram:

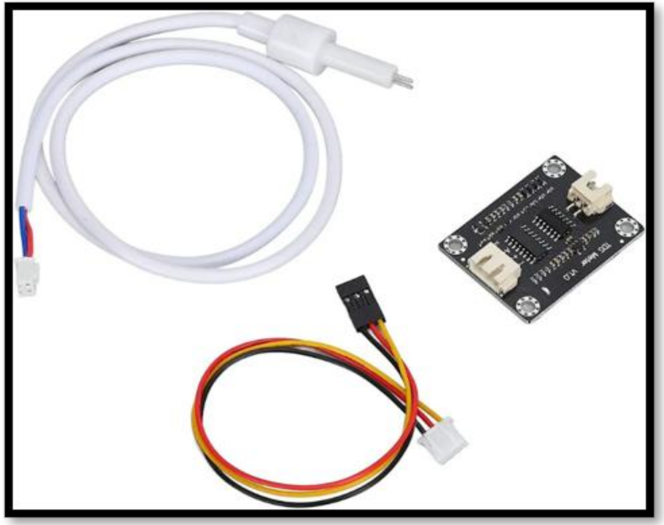


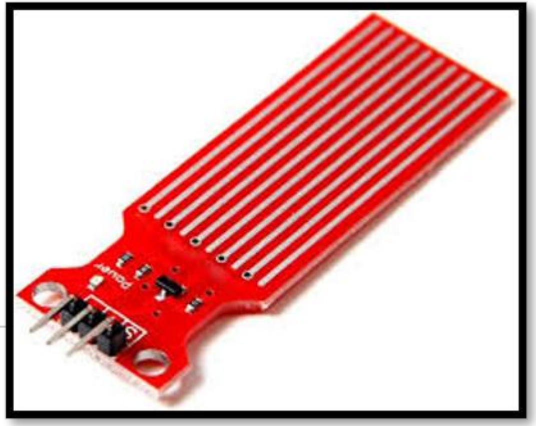
Working

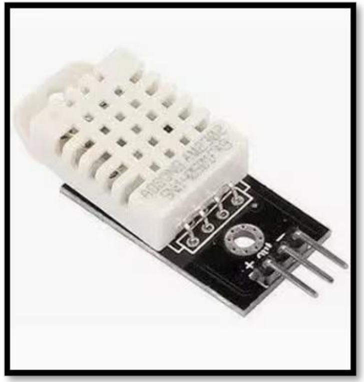

START
Initialize System
Read Sensor Data
Send Data to Blynk
YES
Is Water Level Low?
NO
Send Notification on Blynk App
Keep Pump OFF
Send Email Alert
Turn ON Pump
Check Parameters (pH, Temp, TDS)
YES
Are Parameters within range?
NO
Send Data to Blynk
Send Alert
Display values on OLED
Repeat Loop

Hardware components:





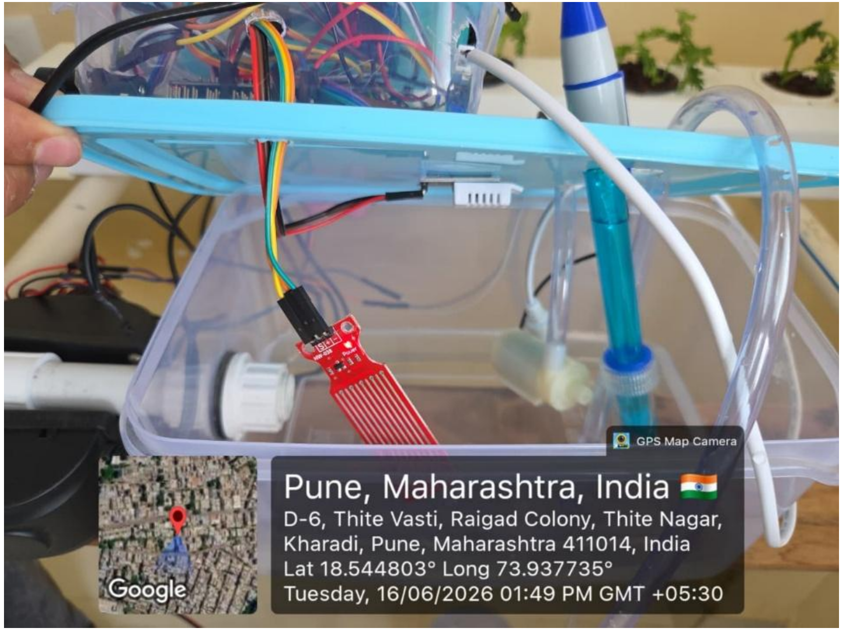




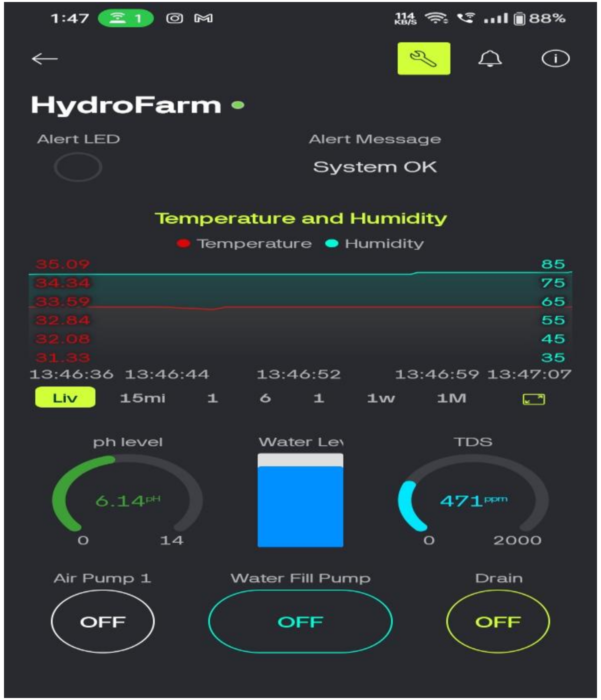
Software requirements:

- •Arduino IDE (for coding ESP32)
- •Blynk IoT App (for mobile dashboard)
- •Proteus

Hardware Implementation



Blynk Dashboard Results



Real-Time Monitoring

Monitored Parameters:

- pH Level
- TDS Value
- Temperature
- Humidity
- Water Level

Features

- Live data visualization
- Remote access
- Wireless monitoring

Notification and Alert

Low Water Level Alert

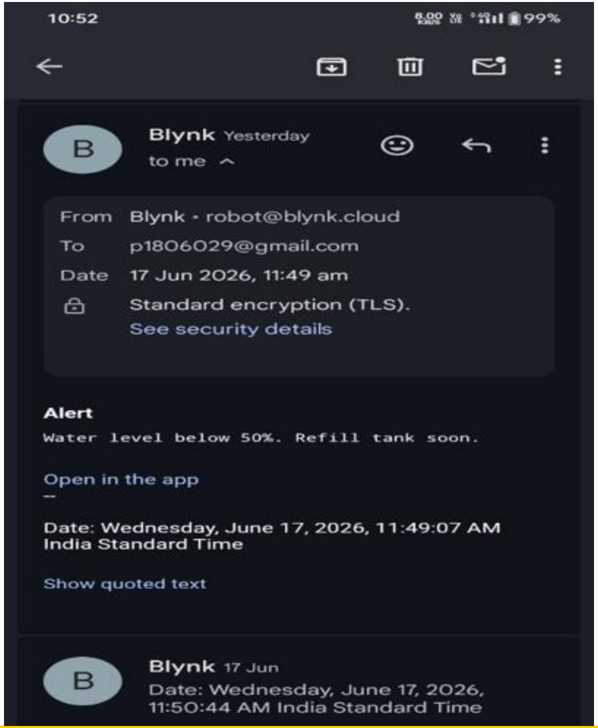
When water level falls below threshold:

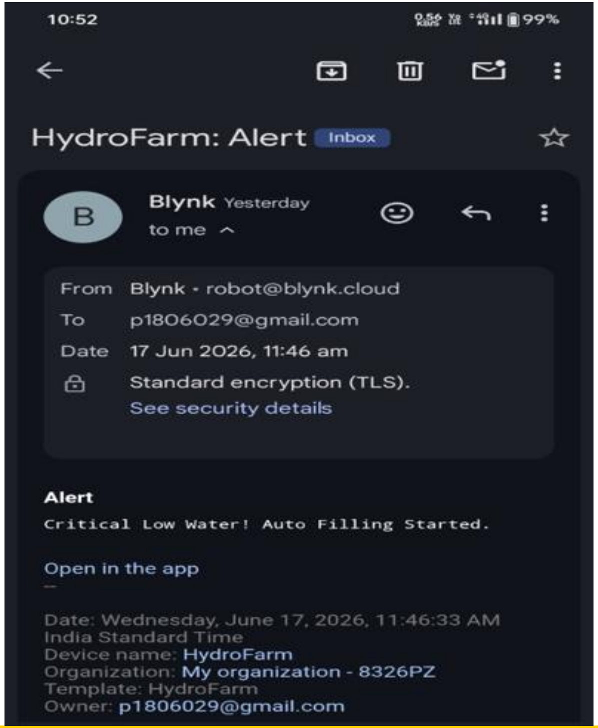
- Blynk push notification generated.
- Email alert sent to user.
- Immediate action can be taken.

Benefits

- Prevents dry operation.
- Improves reliability.
- Enables remote supervision.

Email Alert


10:52
99%
Blynk Yesterday
to me
From Blynk • robot@blynk.cloud
To p1806029@gmail.com
Date 17 Jun 2026, 11:49 am
Standard encryption (TLS).
See security details
Alert
Water level below 50%. Refill tank soon.
Open in the app
--
Date: Wednesday, June 17, 2026, 11:49:07 AM
India Standard Time
Show quoted text
Blynk 17 Jun
Date: Wednesday, June 17, 2026,
11:50:44 AM India Standard Time


10:52
99%
HydroFarm: Alert Inbox
Blynk Yesterday
to me
From Blynk • robot@blynk.cloud
To p1806029@gmail.com
Date 17 Jun 2026, 11:46 am
Standard encryption (TLS).
See security details
Alert
Critical Low Water! Auto Filling Started.
Open in the app
--
Date: Wednesday, June 17, 2026, 11:46:33 AM
India Standard Time
Device name: HydroFarm
Organization: My organization - 8326PZ
Template: HydroFarm
Owner: p1806029@gmail.com

Experimental Results

Parameter	Ideal Values
pH Level	5.5-6.8
Temperature	23- 35°C
Humidity	50 – 70 %
TDS Value	400 – 900 ppm
Water Level	Normal

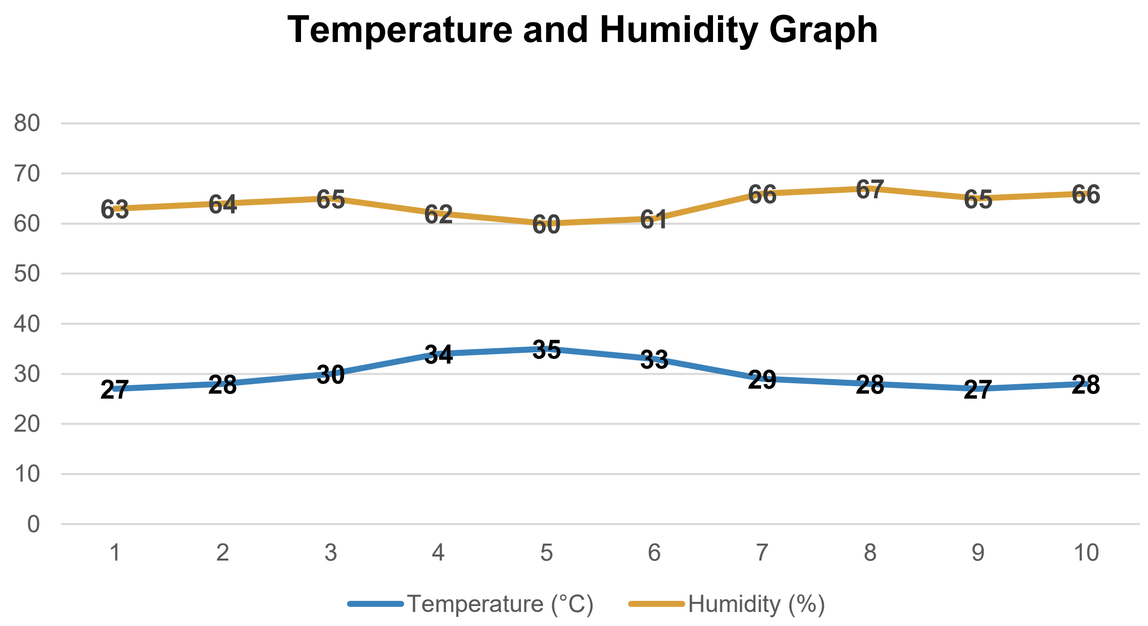
Experimental Results

Day	pH	TDS (ppm)	Temperature (°C)	Humidity (%)	Water Level	Remarks
1	6.2	710	27	63	Normal	Normal
2	6.3	715	28	64	Normal	Normal
3	6.4	720	30	65	Normal	Normal
4	6.3	725	34	62	Normal	High Ambient Temp
5	6.4	718	35	60	Low	Alert Generated
6	6.2	722	33	61	Normal	Stable
7	6.3	730	29	66	Normal	Normal
8	6.4	725	28	67	Normal	Normal
9	6.3	720	27	65	Normal	Normal
10	6.4	728	28	66	Normal	Normal

Graphical Analysis

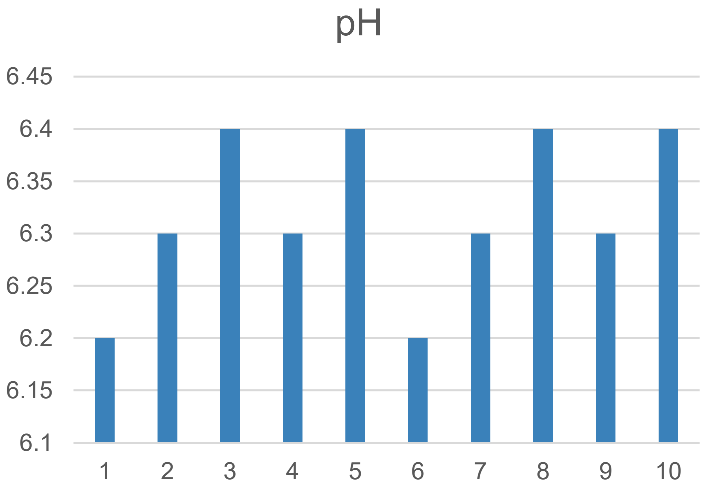
Observation

- Stable environmental conditions maintained.
- Continuous monitoring achieved.
- Reliable sensor performance observed.

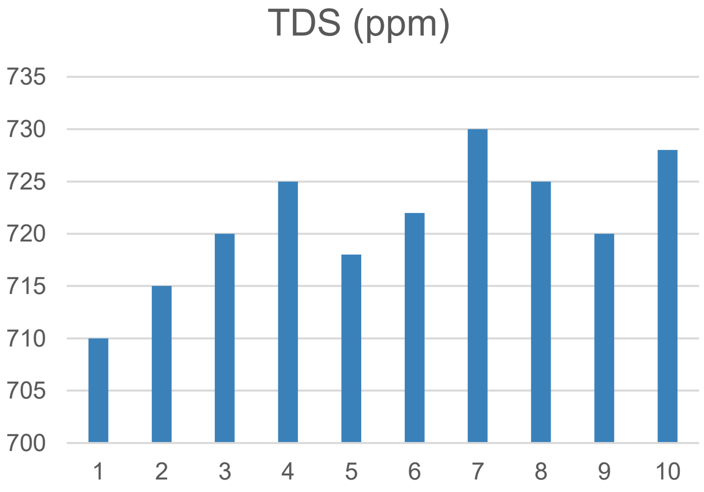


Graphical Analysis

pH graph



TDS Graph



Results Obtained

Successfully Achieved

- ✅ Real-time monitoring
- ✅ Wi-Fi communication
- ✅ Blynk integration
- ✅ OLED display monitoring
- ✅ Low water level detection
- ✅ Blynk notifications
- ✅ Email alerts
- ✅ Automated refill operation

Conclusion and Future Scope

Conclusion

- System implemented successfully
- Reliable monitoring achieved
- Reduced manual intervention
- Improved resource utilization

Future Scope

- AI-based prediction
- Cloud analytics
- Solar-powered operation
- Large-scale deployment

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Thanks!